

MMD Generative Adversarial Networks

Meeting 3

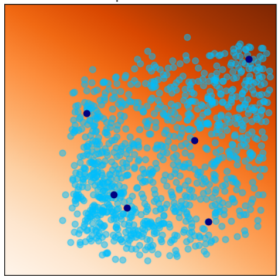
June 10, 2026

- 1 Analysis of stability of MMDGAN Training in 2d examples , Hyperparameters effects
- 2 Kernel choice effect
- 3 Convexity of the target distribution
- 4 High dimension initialisation

Different values of σ in the RBF Kernel

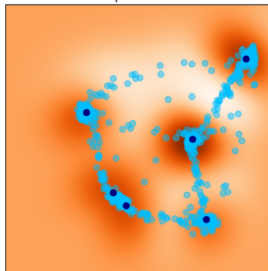
$\sigma = 2$

Epoch 295



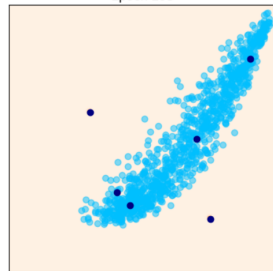
$\sigma = 0.1$

Epoch 295



$\sigma = 0.005$

Epoch 295



The MMD/Kernel Gradient

In an MMD GAN, the generator updates its parameters based on the gradient of the MMD distance. For a Gaussian Radial Basis Function (RBF) kernel:

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

The gradient of this kernel with respect to a generated point x scales directly with the bandwidth parameter σ :

$$\nabla_x k(x, y) = -\frac{1}{\sigma^2}(x - y) \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

Kernels of the type $k(x, y) = f(\|x - y\|^2)$

$$\nabla_x k(x, y) = 2(x - y)f'(\|x - y\|^2)$$

Kernels of the form $k(x, y) = f(||(x - y)||^2)$

Positive definite kernels of the form $k(x, y) = f(||(x - y)||^2)$

A kernel of the form $k(x, y) = f(||(x - y)||^2)$ is positive definite if and only if $f(\mathbf{R}^+) \subset \mathbf{R}_*^+$

f strictly decreasing on R^+

Let $f : R^+ \rightarrow \mathbf{R}_*^+ \in C^2(R^+)$, $f(0) = 1$ strictly decreasing on R^+

We suppose here that $f''(\mathbf{R}^+) \subset \mathbf{R}^+$

We define the d -bandwidth of f as $\sigma = \inf\{x | f'(x) < -d\}$

(Tradeoff bandwidth/Discriminative power)

Under the above assumptions:

$$\sigma d \leq 1$$

The Small Bandwidth Regime ($\sigma \rightarrow 0$)

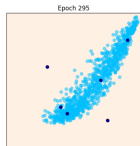
Vanishing gradient due to the small bandwidth

$$\lim_{\sigma \rightarrow 0} \nabla_x k(x, y) = 0 \quad \text{for } x \neq y$$

The generator receives **zero learning signal** guiding it toward the target distribution, completely freezing the learning efficiency.

- **Super-Exponential Decay:** The term $\exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right)$ dominates the gradient calculation and drops to zero at an extreme rate.
- **Flat Optimization Landscape:** The loss surface becomes a collection of isolated plateaus separated by wide, flat valleys.

$\sigma = 0.005$



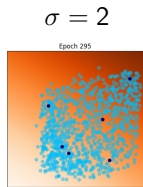
The Large Bandwidth Regime ($\sigma \rightarrow \infty$)

Uniformity caused by the large bandwidth

$$\nabla_x k(x, y) \approx -\frac{1}{\sigma^2}(x - y) \rightarrow 0 \quad \text{as } \sigma \rightarrow \infty$$

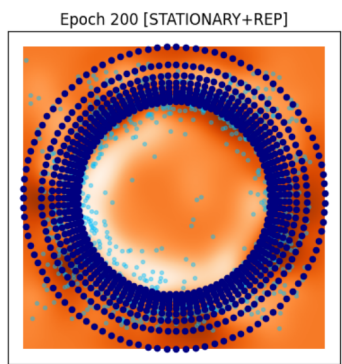
The gradient signal becomes uniform and incredibly weak. The network only learns the global mean, leaving the generator unable to capture complex, high-frequency details.

- **Loss of Discriminative Power:** The exponential term $\exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \rightarrow 1$, making the kernel blind to localized geometric differences.
- **Vanishing Scaling Factor:** The coefficient $\frac{1}{\sigma^2}$ shrinks the magnitude of the global gradient vectors down to near zero.



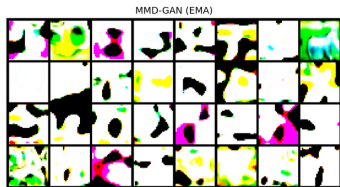
Transporting White noise into a discrete probability measure using a neural network

Using a repulsion force in the stationary phase

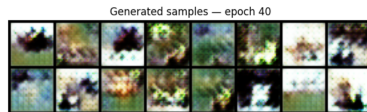


The high dimension initialisation

Computing MMD in a 256 dimensional space



Feature space with $d = 16$



The high dimension initialisation

Relative spread

Let $z_i = x_i - y_i \sim \mathcal{N}(0, 2)$. Then:

$$E[\|\mathbf{x} - \mathbf{y}\|^2] = 2d$$

$$\text{Var}(\|\mathbf{x} - \mathbf{y}\|^2) = 8d$$

Relative standard deviation:

$$\frac{\text{Std}(\|\mathbf{x} - \mathbf{y}\|^2)}{E[\|\mathbf{x} - \mathbf{y}\|^2]} = \frac{\sqrt{8d}}{2d} = \sqrt{\frac{2}{d}} \xrightarrow{d \rightarrow \infty} 0$$

Relative spread $\sqrt{2/d}$ vs dimension

dim d	$\sqrt{2/d}$	Signal
16	0.354	Strong
64	0.177	Moderate
256	0.088	Weak
384	0.072	Collapsed

Why the MMD initialisation Collapses in high dimension

Exact MMD^2 for Gaussian shift

Let $P = \mathcal{N}(0, I_d)$, $Q = \mathcal{N}(\mu e_1, I_d)$ with $\mu > 0$, $t = 1/(2\sigma^2)$. Then:

$$\text{MMD}^2(P, Q) = 2(1 + 4t)^{-d/2} \left[1 - \exp\left(-\frac{t\mu^2}{1 + 4t}\right) \right]. \quad (1)$$